Traffic Navigation and Congestion Model

CITS4403 Research Project Report

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Abstract

This project builds a reproducible agent-based model to study how traffic demand, navigation-app adoption, and routing policy affect congestion after a road disruption. The report compares selfish navigation with coordinated routing, using static shortest-path routing as the uninformed baseline. The model runs on a small synthetic directed network with capacity-dependent travel times and individual vehicle agents, and it comes with seeded experiments, automated tests, and generated figures. Selfish navigation can improve travel time at intermediate adoption levels, but the benefit shrinks at full adoption. Coordinated routing gives the lowest mean journey time in the scenarios tested. These findings apply to the synthetic model and its assumptions, not to real traffic conditions.

1 Background and research aim

Congestion is partly the combined result of many individual route choices. A driver may pick what looks like the fastest route, but that choice can add delay for other drivers on the same road. Wardrop's first principle separates decentralised user-equilibrium choices from a system-optimal assignment that minimises total network cost [5]. The distinction matters for navigation apps, which generally recommend whatever route suits a single trip.

Braess's paradox points the same way. Under some cost functions, adding a route or giving drivers more information can raise the equilibrium travel cost [3, 2]. Recent work has also treated navigation-app adoption as an experimental factor. One simulation study found that intermediate adoption of dynamic routing could outperform both low and very high adoption when rerouting spreads traffic across a network [4]. A field experiment reported congestion relief after a small share of trips was redirected away from selected road segments [1].

None of these studies shows how the small synthetic network used here behaves. This project therefore tests the same general mechanism under controlled assumptions.

Research question. How do traffic demand and real-time navigation-app adoption affect system-wide congestion and travel times after a road disruption?

The hypothesis is that navigation reduces travel time at low or moderate adoption. If many drivers receive the same information and choose the same alternative route, though, widespread adoption may overload those roads and reduce the benefit.

2 Originality and contribution

The main contribution is a small, inspectable experiment that combines individual vehicle agents, capacity-constrained road costs, and an external disruption. It extends a static shortest-path example by making the proportion of agents with current traffic information an explicit

experimental variable. The synthetic topology is deliberate. It makes the feedback between route choice and bottlenecks easier to observe, and it allows boundary cases to be tested without the complexity of a real road map.

The project also separates personal and collective objectives through two research policies. Static routing, the baseline, uses shortest paths based on free-flow travel times. Selfish routing gives app users current travel-time information and optimises a personal objective. Coordinated routing evaluates routes against a network-wide objective. The report focuses on the difference between decentralised route choice and coordination at the system level.

3 Model specification and justification

The road network is a directed graph $G = (V, E)$. Nodes represent intersections and directed edges represent road segments, with separate edges for opposite directions. The network has six nodes arranged as two parallel corridors between an origin-destination region (Figure 1). Each edge stores its free-flow travel time $t_{0,e}$, capacity C_e, occupancy x_e, and current travel time t_e. Time is discrete and measured in simulation steps, while occupancy and capacity are measured as vehicle counts.

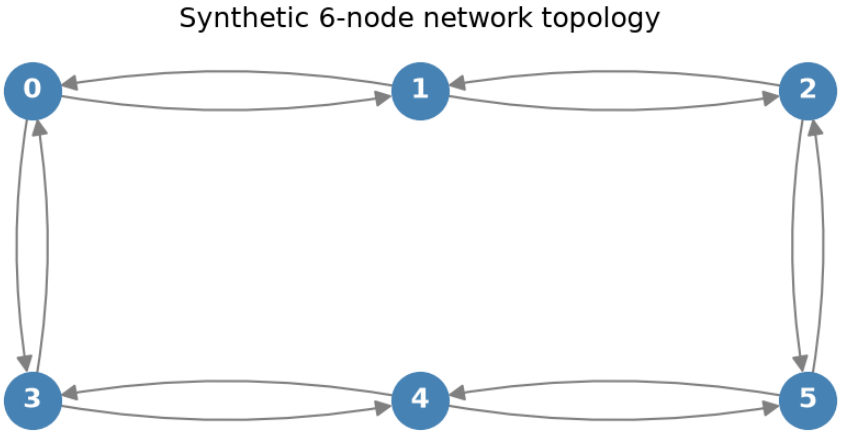


Figure 1: Synthetic six-node network design. The two corridors provide alternative paths while keeping the route-choice mechanism straightforward to inspect.

Congestion follows a Bureau of Public Roads-style volume-delay function:

$$t_e = t_{0,e}\left[1 + \alpha\left(\frac{x_e}{C_e}\right)^{\beta}\right]. \tag{1}$$

The parameters are $\alpha = 0.15$ and $\beta = 4.0$. These are modelling choices, not empirical estimates. The equation is monotonic in occupancy, gives free-flow travel time when occupancy is zero, and rises sharply as demand approaches or exceeds capacity (Figure 2). A closed road is excluded from newly planned routes or represented using zero effective capacity.

Each vehicle has a unique identifier, origin, destination, route, current position, start time, and completion time. Static vehicles use minimum free-flow paths found with Dijkstra's algorithm. Selfish vehicles use current edge costs, and coordinated routing evaluates assignments against a network-wide objective. At each time step, a vehicle enters the network, moves along its route, transfers to the next edge, or completes its journey. Occupancies are then updated and used to

refresh travel times. Routes are fixed at entry, so the experiment models route choice at the start of a journey, not continuous mid-trip rerouting.

The model leaves out lane changing, traffic signals, collisions, parking, pedestrians, and real map geometry, so that the experiment isolates the relationship between capacity, disruption, and route information. Randomness is limited to demand generation and origin-destination assignment, and each experiment records its configuration and explicit seed.

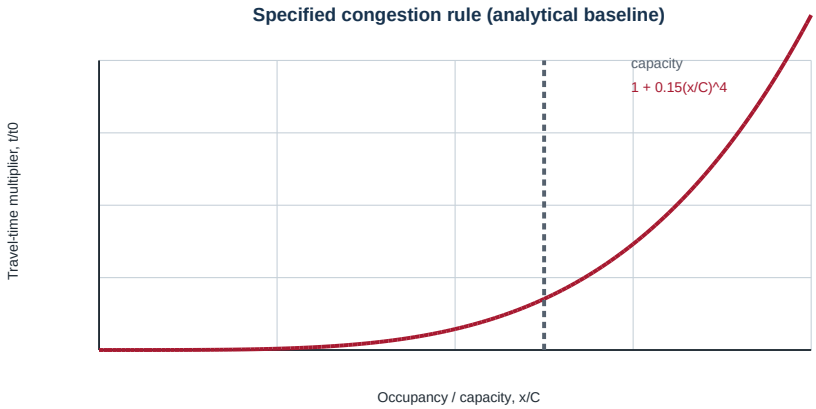


Figure 2: Analytical implication of the selected congestion rule. Travel time does not decrease as the occupancy-to-capacity ratio increases.

4 Experimental design

The focused comparison uses two research policies, five adoption rates (0%, 25%, 50%, 75%, and 100%), two disruption states, and five fixed seeds (0 to 4). This gives 100 summary rows: $2 \times 5 \times 2 \times 5$. Static routing remains the reference baseline in the broader routing-comparison output. For selfish routing, which depends on adoption, the 0% condition is the uninformed case.

The broader experiment also varies demand level and disruption type. Which of these appear in the final summary must match the experiment scripts and the reported row count.

Factor	Levels	Purpose
Demand	Low / medium / high	Test the effect of increasing network load
Adoption	0 / 25 / 50 / 75 / 100%	Test the effect of navigation information
Disruption	None / reduction / closure	Test robustness and recovery
Replication	Multiple fixed seeds	Measure stochastic variation

Table 1: Planned experiment factors and their purpose.

Each raw summary records the scenario parameters, seed, completed journeys, mean, median, and 95th-percentile journey time, total network delay, peak congestion, and disruption-recovery measures. The congestion output also records road occupancy at each timestep. Automated tests check route validity, monotonic congestion, closed-road exclusion, disruption behaviour, seeded adoption, repeatability, and policy-specific outcomes. The route checks confirm that every route starts at its origin and ends at its destination.

5 Results and current evidence

The repository contains the runnable Python source in `src/`, experiment scripts in `scripts/`, tests in `tests/`, the final analysis notebook, and raw CSV outputs in `results/`. The test suite has 220 passing tests. The focused two-policy CSV has 100 rows covering the two research policies, five adoption rates, two disruption states, and five seeds.

The figures below are generated experiment outputs. They should be read alongside the CSV files and the fixed-seed procedure, not as standalone screenshots.

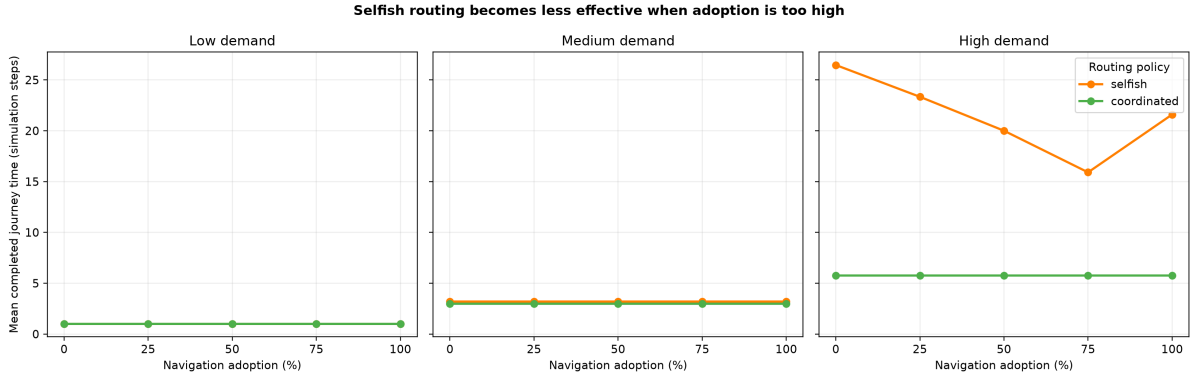


Figure 3: Mean completed journey time across navigation adoption and demand. Selfish and coordinated routing are compared using the same seeded scenario grid.

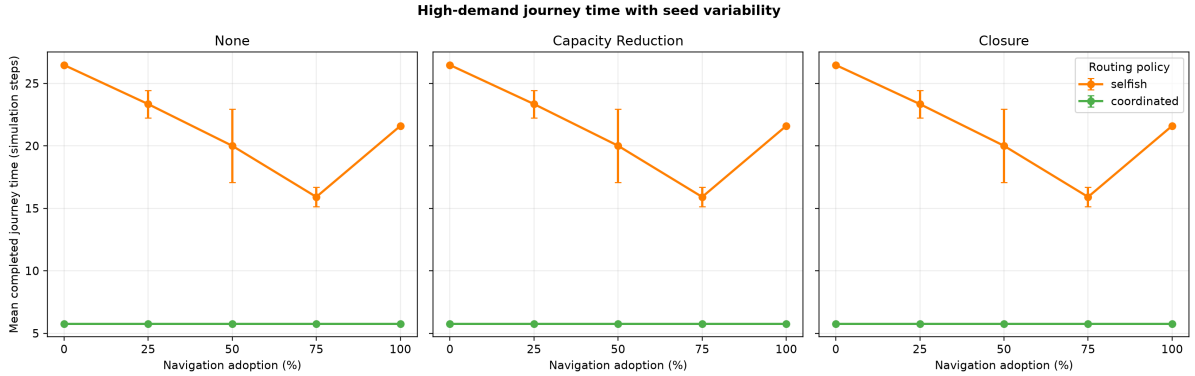


Figure 4: High-demand journey time across seeds under the two research policies. Points and error bars show the variation retained across repeated runs.

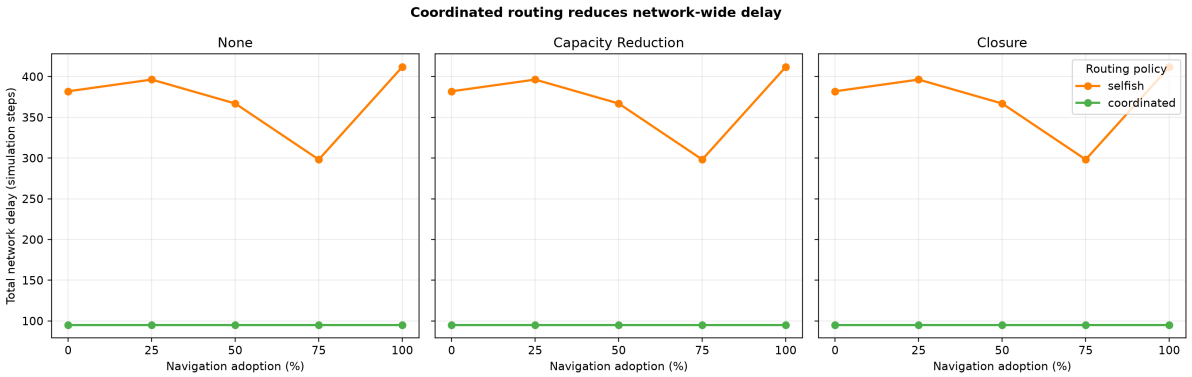


Figure 5: Mean network congestion after capacity reduction and closure. Time is measured in simulation steps; dashed lines show disruption onset and dotted lines show restoration.

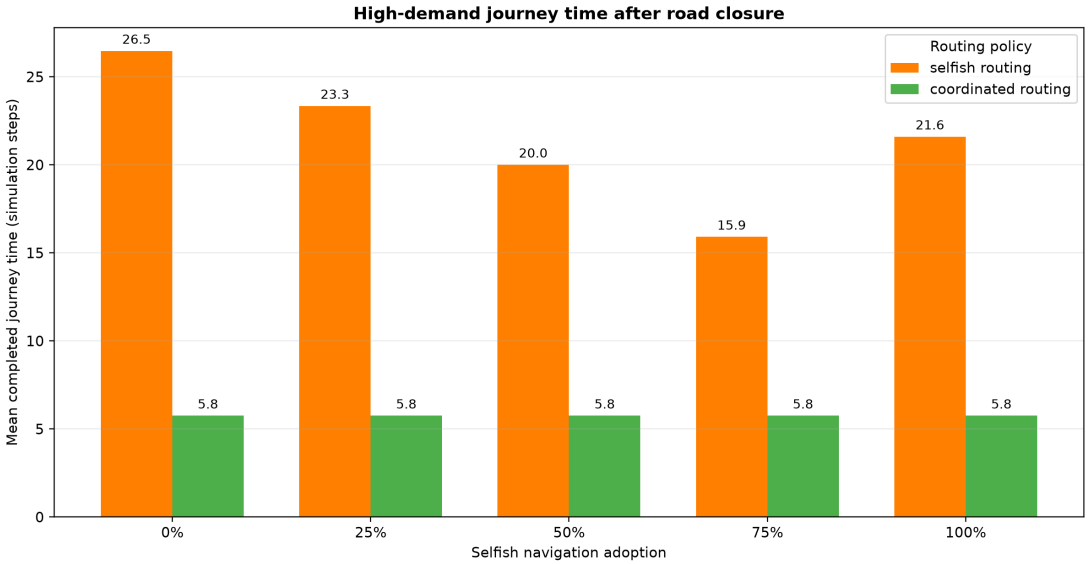


Figure 6: Selfish-navigation advantage relative to static routing. Positive values indicate lower mean journey time, measured in simulation steps.

Table 2 gives the BPR multipliers, which serve as the verified quantitative baseline. The multiplier is monotonic, as expected: increasing occupancy does not reduce travel time. Under static routing, mean journey time in the generated demand validation also rises, from 1.0 steps at low demand to 2.4 steps at medium demand and 2.8 steps at high demand.

Occupancy ratio x/C	Multiplier t/t_0	Interpretation
0.0	1.000	Free flow
0.5	1.009	Small delay under light load
1.0	1.150	Capacity reached
1.5	1.759	Over-capacity state

Table 2: Deterministic baseline values from Equation 1, calculated as $1 + 0.15(x/C)^4$.

6 Discussion, limitations, and conclusion

The results support the hypothesis, but only under the conditions tested. At high demand with no disruption, static routing has a mean journey time of 26.47 steps and completes 75% of vehicles. Selfish navigation lowers the mean journey time to 15.91 steps at 75% adoption, but the mean journey time rises again to 21.60 steps at 100% adoption. Coordinated routing has a mean journey time of 5.75 steps and completes all vehicles in these high-demand comparisons. The non-monotonic response from selfish routing is consistent with correlated route choice: many informed vehicles may concentrate on the same alternative road.

The disruption figure shows the same difference between personal and system-level objectives. Selfish navigation reduces congestion more quickly than static routing after both capacity reduction and closure, and coordinated routing gives the strongest system-level benchmark. None of this shows that one policy is always best. Coordinated routing may assign different routes to individuals, so completion rate, individual journey time, and network congestion need to be considered together.

The synthetic network limits the external validity of the findings. The BPR parameters, demand process, and driver responses are simplified assumptions. Five seeds provide repeatability and a first indication of variation, but they are not a confidence interval, and the model is not calibrated against observed traffic. Routes are planned at entry rather than continuously rerouted, so the model does not reproduce every behaviour of a commercial navigation app. In the comparison runner, selfish routing uses a pre-entry traffic snapshot, a deliberate simplification of live information.

Overall, the project is a reproducible test of how demand, navigation adoption, disruption, and routing objectives interact in a capacity-constrained network. Selfish navigation helps under some conditions, coordination gives a stronger system-level outcome, and decentralised routing is penalised at high adoption. These conclusions apply only to the synthetic network and its assumptions, and they would need calibration against real-world data before they could say anything about actual Perth traffic.

References

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